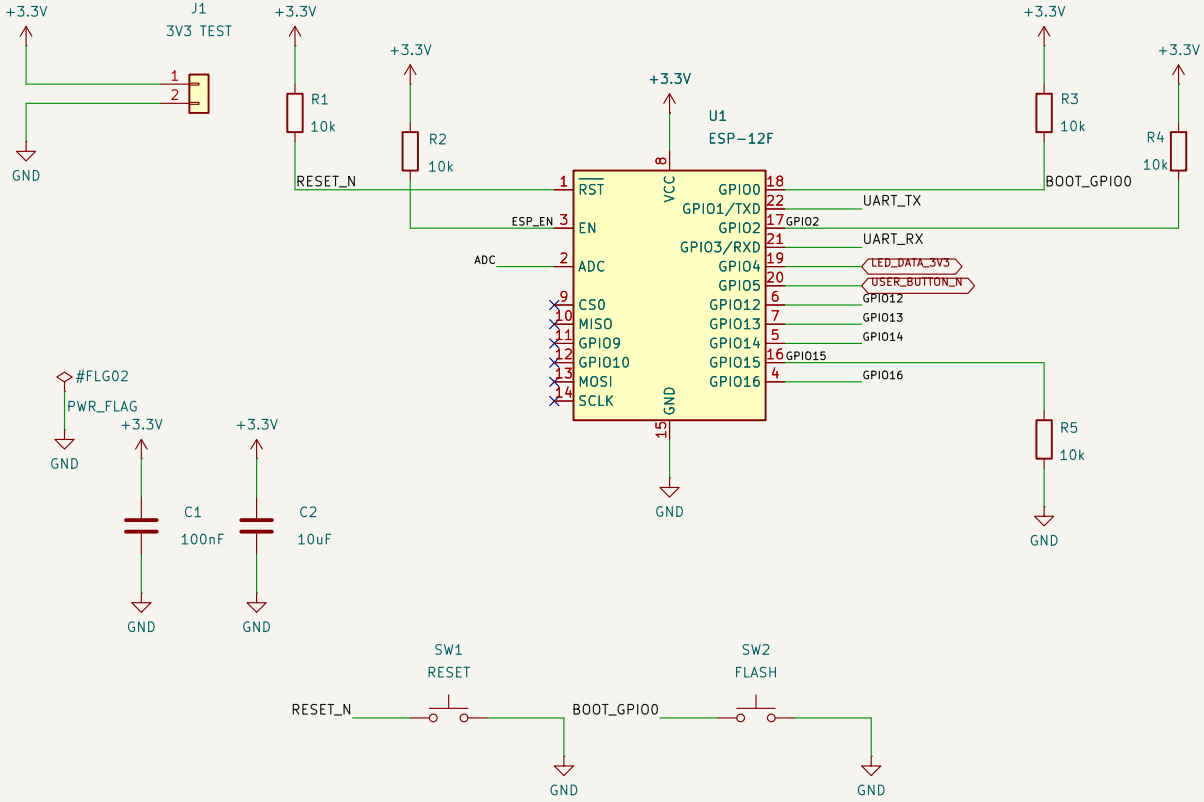
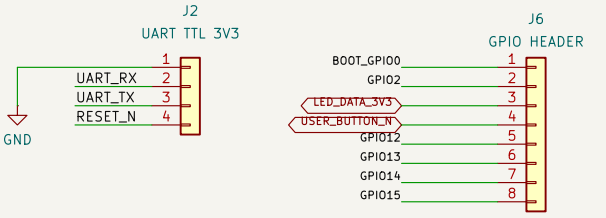
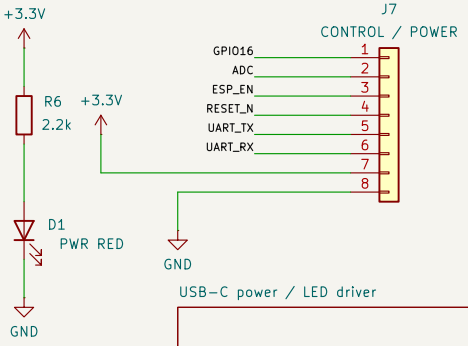
RGB Sleep Lamp — ESP-12F controller

J1: 3.3V test header. Supplied from power sheet; do not feed from UART adapter.







USB-C power / LED driver

File: power.kicad_sch

MP1584 discrete buck

File: buck.kicad_sch

J2: pin 2 <- adapter TX; pin 3 -> adapter RX. Common GND.

UART logic 3.3V only. Real RS-232 requires a level converter.

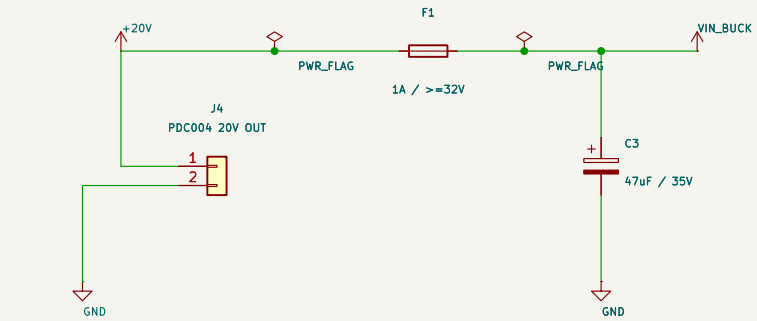
Upload: hold FLASH, pulse RESET, release FLASH, upload. Then RESET.

Place C1/C2 near module VCC/GND. ADC: 0-1V maximum at module pin.

PWR LED: red low-current type; ~0.6mA nominal (Vf=2.0V).

Sheet: /		
File: circuit.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/3

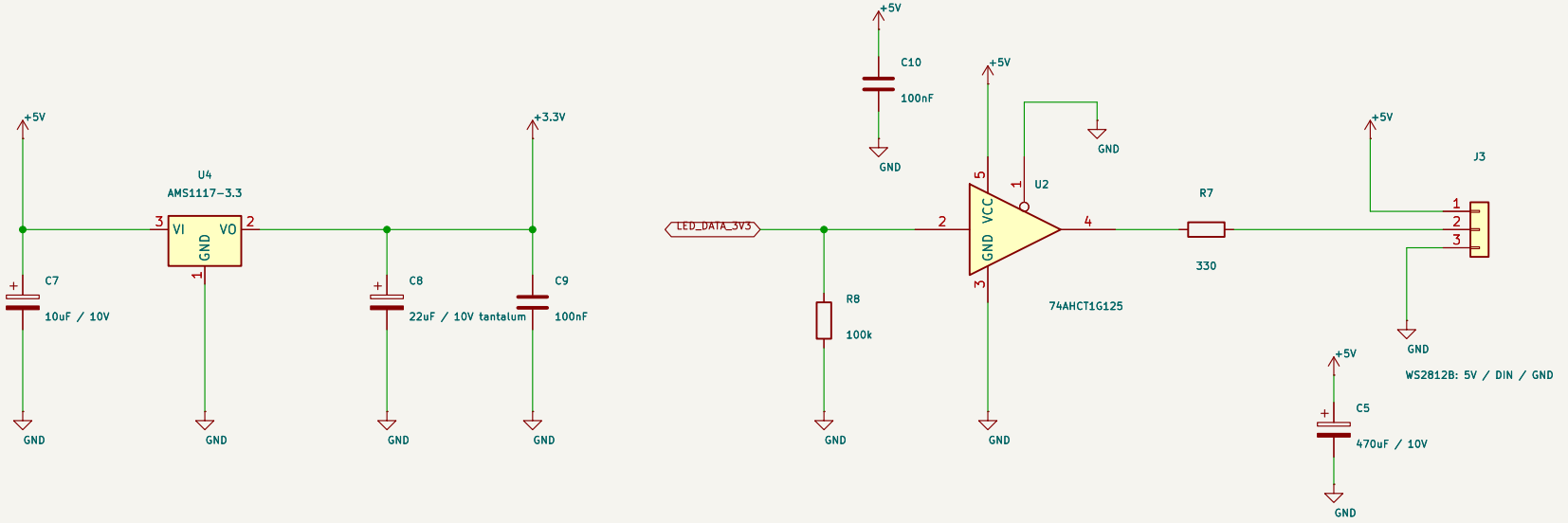
USB-C INPUT / 3V3 REGULATOR / LED DRIVER

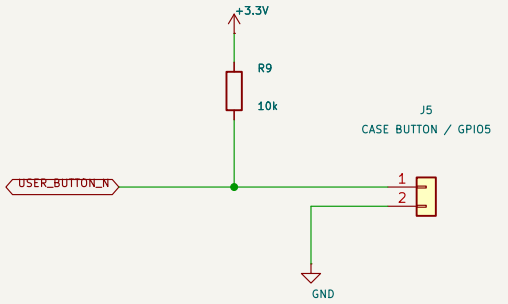




J4: solder two wires from PDC004 OUT+ / OUT−. USB-C remains on PDC004.

MP1584EN + donor components are on buck.kicad_sch; VIN_BUCK is the fused 20V rail.





30 LEDs: budget 1.8A at 5V + ESP allowance up to 0.5A. Target 5V/2.5A.

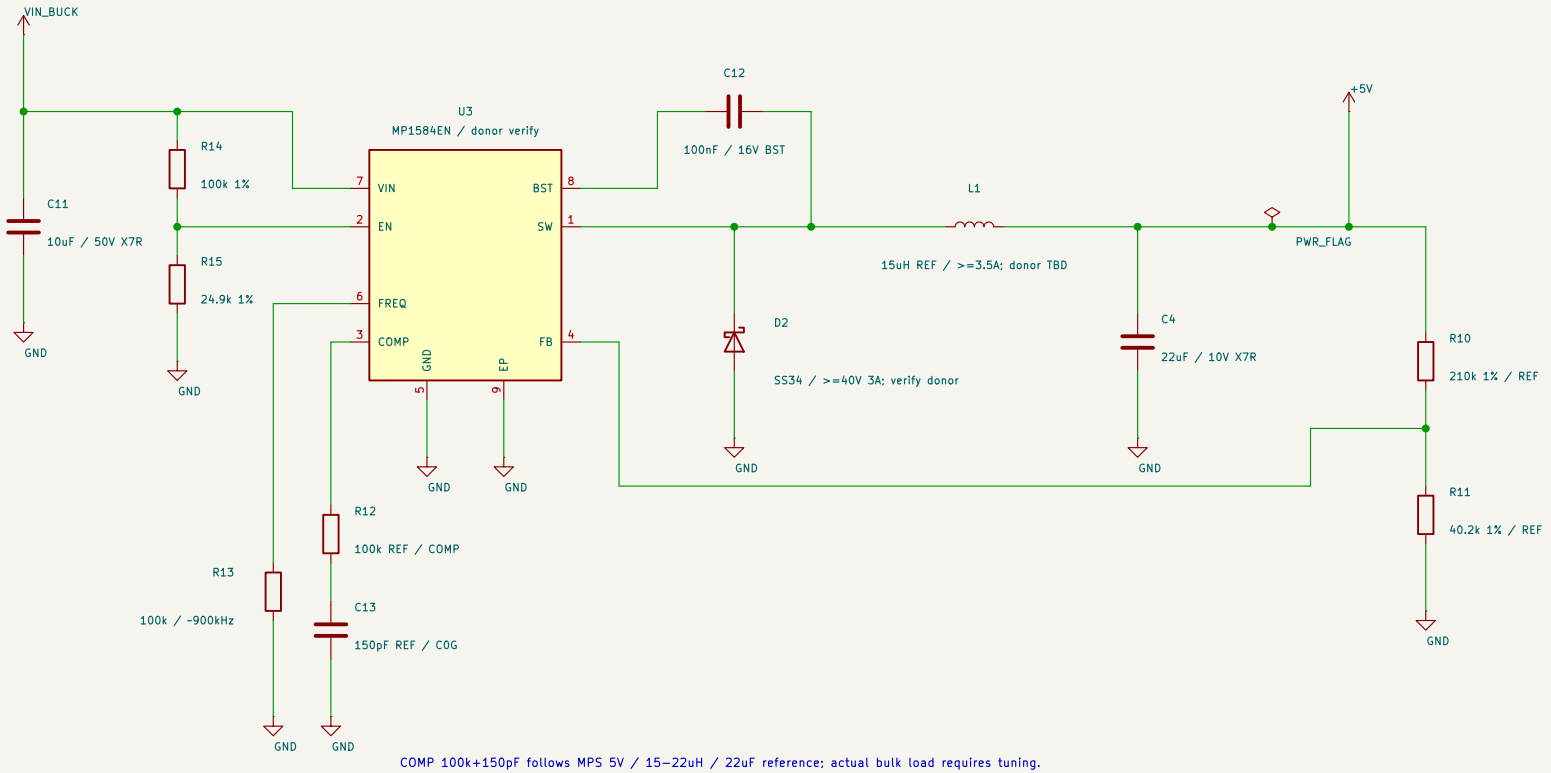
Power rating is provisional: verify buck temperature/ripple in closed enclosure.

GPIO4 = LED data; GPIO5 = case button. Header pins share these functions.

UART is 3.3V TTL; PDC004 USB-C is POWER ONLY, no USB data/programming.

Sheet: /USB-C power / LED driver/ File: power.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/3

MP1584EN DISCRETE BUCK — SALVAGED COMPONENTS ON MAIN PCB



COMP 100k+150pF follows MPS 5V / 15–22uH / 22uF reference; actual bulk load requires tuning.

C5 470uF is on power/LED sheet: validate total output ESR, startup, loop stability and full-white load.

EN divider inhibits low input; EP pad9 must join GND. Short VIN–C11–U3–D2 and SW–D2–L1 loops.

Reference 5V values: Rfb 210k/40.2k -> 4.979V nominal; verify donor IC marking and all values.

L1 / resistor / capacitor values and footprints are reference placeholders until donor measurement.

Sheet: /MP1584 discrete buck/ File: buck.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/3